Graduate Seminar 2

26121301

Ongonwou Fred

Phd Candidate
2343003

May 27, 2026

Graduate Seminar 2

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Section 1 - Solving for $\psi(x,t)$

- We solve $\psi(x,t)$ by a numerical resolution of teh Time-Dependent Schödinger Equation (TDSE),i.e

$$i\frac{\partial}{\partial t}\psi(x,t) = \left(\hat{H}_a + \hat{V}_{\text{ext}}(t)\right)\psi(x,t), \qquad \psi(x,0) = \psi_0(x), \tag{1}$$

$$\hat{H}_a = -\frac{1}{2}\frac{d^2}{dx^2} - \varkappa\delta(x), \qquad \hat{V}_{\text{ext}}(t) = F(t)x. \tag{2}$$

$$E_0 = -\varkappa^2/2, \qquad \psi_0(x) = \varkappa^{1/2}e^{-\varkappa|x|}, \qquad \varkappa = 1, \tag{3}$$

$$E_k = k^2/2, \qquad \psi_k^{(\pm)}(x) = e^{ikx} + R(\pm|k|)e^{\pm i|kx|}, \qquad R(k) = \frac{i\varkappa}{k - i\varkappa}. \tag{4}$$

- The parameters of the laser we will be using are given by the relations

$$F(t) = F_0 e^{-(2t'/\tau)^2}\cos(\omega_0 t'), \qquad 0 \leqslant t \leqslant T, \qquad t' = t - T/2, \tag{5}$$

$$\tau = 2\pi n_{oc}/\omega_0, \qquad n_{oc} = 5. \tag{6}$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Density, survival probability and PEMD

- The density probability of the wavefunction, at t is given by

$$\mathcal{D}(x,t) = |\psi(x,t)|^2 = \int \psi^*(x,t)\psi(x,t)dx. \tag{7}$$

- The coefficients of the projection of ψ are then, at t, and by definition, expressed as

$$a_0 = e^{iE_0 t}\int \psi_0(x)\psi(x,t)dx, \qquad a_k = e^{iE_k t}\int \psi_k^{(-)*}(x)\psi(x,t)dx. \tag{8}$$

- Then,

$$P_0 = |a_0|^2, \qquad \text{and} \qquad P(k) = |a_k|^2. \tag{9}$$

- The ionization probability can then be found many different ways that should be consistent with each others

$$P_{\text{ion}} = \int P(k)\frac{dk}{2\pi} = 1 - P_0. \tag{10}$$

- $P(k)$ is the probability for the electron being ionized with the energy E_k.

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Observables

- For an arbitrary operator $\mathcal{O}$, we have

$$\mathcal{O}(t) = \int \psi^*(x,t)\mathcal{O}\,\psi(x,t)dx \tag{11}$$

- The the Fourier transform is obtained using

$$\mathcal{O}(\omega) = \int \mathcal{O}(t)e^{i\omega t}dt. \tag{12}$$

- Following Bransden an Joachain, "Quantum Mechanics (2000)" (see 11.145), we define the transition amplitude $\mathcal{O}_{ba}$ and probability P_{ba} between two states a and b such as

$$\mathcal{O}_{ba} = \int dx\,\psi_b^*(x)\,\mathcal{O}\,\psi_a(x), \qquad P_{ba} = |\mathcal{O}_{ba}|^2\,. \tag{13}$$

When continuum states between q and $q+dq$ are involved, the quantity of interest is

$$P_{ba}dq = \left|\int dx\,\psi_b^*(x)\,\mathcal{O}\,\psi_a(x)\right|^2 dq. \tag{14}$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Length, velocity and acceleration

- We want to evaluate the quantities

$$d(t) = -\langle \hat{x} \rangle = -\int \psi^*(x,t)\hat{x}\,\psi(x,t)dx, \qquad \text{and} \qquad \dot{x}(t) = \frac{d}{dt}x(t), \tag{15}$$

$$p(t) = \langle \hat{p} \rangle = -i\int \psi^*(x,t)\frac{\partial}{\partial x}\psi(x,t)dx. \tag{16}$$

- Direct numerical differentiation can sometimes be avoided with the help of the Ehrenfest theorem and the commutator algebra

$$\left[\hat{p}^{\,2}, \hat{x}\,\right] = -2i\hat{p} \quad \Longrightarrow \quad \hat{p} = i\left[\hat{H}\,, \hat{x}\,\right], \tag{17}$$

$$\frac{d}{dt}\langle \hat{x}\,\rangle = \langle \hat{p}\,\rangle, \qquad \frac{d}{dt}\langle \hat{p}\,\rangle = \left\langle -\frac{\partial V_{\text{ext}}}{\partial x} \right\rangle. \tag{18}$$

  and some others (see Bransden and Joachain, Quantum Mechanics (2000), p.531-532)

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

High Harmonic Generation (HHG) - (semi-classical approx.)

- In the classical approx., the spectrum is obtained from the following (Eq.(80))

$$Q_\nu^{(c)}(\omega) = \alpha^2\omega^2 \left|\mathbf{A}_\nu \mathbf{d}(\omega)\right|^2, \tag{19}$$

  Let us apply the following simplifications

$$\nu \longrightarrow \omega, \qquad \mathbf{A}_\nu = \sqrt{\frac{2\pi c^2}{\omega L^3}}\boldsymbol{e}_\nu \longrightarrow A_\omega, \qquad \text{and} \qquad \alpha A_\omega = 1. \tag{20}$$

- So the observable we are considering here is

$$Q^{(c)}(\omega) = \omega^2 \left|d(\omega)\right|^2. \tag{21}$$

  Additionally let us consider Eq. (104)

$$Q^{(c)}(\omega) = \left|p(\omega)\right|^2. \tag{22}$$

  and, by virtue of $\boldsymbol{p}(t) = -\dot{\boldsymbol{d}}(t)$, which reads

$$Q^{(c)}(\omega) = \left|-\dot{d}(\omega)\right|^2. \tag{23}$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Additional note from Ehrenfest theorem

- Using the integration by parts formula, we get

$$p(\omega) = \int_{-T/2}^{T/2} e^{i\omega t} \frac{dx(t)}{dt} dt, \qquad u(t) = x(t), \quad v(t) = e^{i\omega t}, \tag{24}$$

thus

$$p(\omega) = \int_{-T/2}^{T/2} \frac{d}{dt}\Big(e^{i\omega t} x(t)\Big) dt - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt = \Big[e^{i\omega t} x(t)\Big]_{-T/2}^{T/2} - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt, \tag{25}$$

which reduces to (21) up to a constant.

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Transition amplitude

- From Yangaliev *et al.*, we have

$$p_{k0} = \int dx\, \psi_k^{(-)*}(x)\, \hat{p}\, \psi_0(x) = \frac{2k\varkappa^{3/2}}{\varkappa^2 + k^2}, \tag{26}$$

$$p_{kk'} = \int dx\, \psi_k^{(-)*}(x)\, \hat{p}\, \psi_{k'}^{(-)}(x) = 2\pi k \delta(k - k') - \frac{2\varkappa}{k^2 - k'^2 + i\epsilon} \left(\frac{k'|k|}{|k| - i\varkappa} - \frac{k|k'|}{|k'| + i\varkappa} \right). \tag{27}$$

- Using (17), we can rewrite those expressions as

$$p_{k0} = -i\,(E_k - E_0)\, d_{k0}, \tag{28}$$

$$p_{kk'} = -i\,(E_k - E_{k'})\, d_{kk'}. \tag{29}$$

- Those quantities are another way to study convergence, at the structural level

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Part 2 - Solving $\phi_\omega(x,t)$

- We are now interested in solving the following 1D problem with a ZRP

$$\begin{cases} i\frac{\partial}{\partial t}\psi(x,t) &= \left(\hat{H}_a + \hat{V}_{\rm ext}(t)\right)\psi(x,t), \\ i\frac{\partial}{\partial t}\phi_\omega(x,t) &= \left(\hat{H}_a + \hat{V}_{\rm ext}(t)\right)\phi_\omega(x,t) + e^{i\omega t}S(x,t) \end{cases} \tag{30}$$

where

$$\hat{H}_a = -\frac{1}{2}\frac{d^2}{dx^2} - \varkappa\delta(x), \qquad \hat{V}_{\rm ext}(t) = F(t)x, \qquad S(x,t) = \hat{p}\,\psi(x,t) \tag{31}$$

$$\phi_\omega(x,0) = \frac{-i\varkappa^{3/2}}{\omega}\mathrm{sgn}(x)\left(e^{-\varkappa|x|} - e^{-\varkappa(\omega)|x|}\right), \qquad \psi(x,0) = \varkappa^{1/2}e^{-\varkappa|x|}, \tag{32}$$

$$\hat{p} = -id/dx, \qquad \text{and} \qquad \varkappa(\omega) = \sqrt{\varkappa^2 + 2\omega}. \tag{33}$$

$$\phi_\omega(x,t) = U(t,t_0)\phi_\omega(x,t_0) - i\int_{t_0}^{t} U(t-\tau)e^{i\omega\tau}S(x,\tau)d\tau. \tag{34}$$

High-order exponential splitting (composition methods) (Phys. Rev. A 101, 013410 (2020))

Yoshida (1990) - 4th-order symmetric decomposition

$$\phi_\omega(x, t + \Delta t) = U_4(t + \Delta t, t)\phi_\omega(x, t) - i \int_t^{t+\Delta t} U_4(t + \Delta t - \tau)e^{i\omega\tau} S(x, \tau)d\tau. \tag{34}$$

- From the following Strang operator $U(h) = e^{-iE_n h/2} e^{-iV_{\text{ext}}(t+h/2)h} e^{-iE_n h/2}$, we compose

$$U_4(h) = U(c_1 h)U(c_2 h)U(c_1 h), \qquad \text{with} \qquad c_1 = \frac{1}{2 - 2^{1/3}}, \quad c_2 = -\frac{2^{1/3}}{2 - 2^{1/3}}. \tag{35}$$

- More precisely, we can observe the following

Theorem 8.6 (Hairer, Solving ODE II p.554)
If Ψ_h represent a one-step method that is symmetric, of order $p = 2k$ and defined in the domain, then the relations

$$2c_1 + c_2 = 1, \qquad 2c_1^{p+1} + c_2^{p+1} = 0, \tag{36}$$

imply that the composition (35) is of order $p + 2$.

If you start from the Strang method, then in the above $k = 1$.

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Midpoint Duhamel - 2nd order approximation

$$\phi_\omega(x, t+\Delta t) = U(t+\Delta t, t)\phi_\omega(x,t) - i\int_t^{t+\Delta t} U(t+\Delta t-\tau)e^{i\omega\tau}S(x,\tau)d\tau. \tag{34}$$

- We can show that the above can be expressed as

$$\phi_{n+1}^\omega = U(\Delta t)\phi_n^\omega - i\Delta t\, e^{i\omega t_{n+1/2}} S_{n+1/2}, \tag{37}$$

where the source is evaluated at $t_{n+1/2} = t + \Delta t/2$, i.e and $S_{n+1/2} = U(\Delta t/2)S_n$, with $S_n = S(x, t_n)$

- This is the midpoint Duhamel formula, of order 2.
- It's what will allow us to get familiar with the numerical intricacies of our problem
- The driver shall be regarded as decent if the expected order is met

Geometric propagators

Simpson Duhamel - 4th-order quadrature

$$\phi_\omega(x, t + \Delta t) = U_4(t + \Delta t, t)\phi_\omega(x, t) - i \int_t^{t+\Delta t} U_4(t + \Delta t - \tau)e^{i\omega\tau} S(x, \tau)d\tau. \tag{34}$$

- The simpson quadrature consists in writing

$$\phi^\omega_{n+1} = U_4(\Delta t)\phi^\omega_n - \frac{i\Delta t}{6} \left(U_4(\Delta t)e^{i\omega t_n} S_n + 4U_4(\Delta t/2)e^{i\omega t_{n+\frac{1}{2}}} S_{n+\frac{1}{2}} + U_4(0)e^{i\omega t_{n+1}} S_{n+1} \right) \tag{38}$$

where the source is evaluated at $t_n,\ t_{n+\frac{1}{2}}$ and t_{n+1}

- This is the simpson Duhamel formula, of order 4.
- It requires a 4th order evolution operator to work
- Simpson-Duhamel multiplies the number of operations required by Midpoint-Duhamel by 3.

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Observables

$$b_{0\omega}(T) = e^{iE_0T} \int \psi_0(x)\phi_\omega(x,T)dx, \tag{39}$$

$$b_{k\omega}(T) = e^{iE_kT} \int \psi_k^{(-)*}(x)\phi_\omega(x,T)dx, \tag{40}$$

$$b_{0\omega} = b_{0\omega}(T) + \int \frac{e^{i(E_0+\omega-E_{k'})T} p_{0k'} a_{k'}}{E_0+\omega-E_{k'}+i0} \frac{dk'}{2\pi} \tag{41}$$

$$b_{k\omega} = b_{k\omega}(T) + \frac{e^{i(E_k+\omega-E_0)T} p_{k0} a_0}{E_k+\omega-E_0} + \int \frac{e^{i(E_k+\omega-E_{k'})T} p_{kk'} a_{k'}}{E_k+\omega-E_{k'}+i0} \frac{dk'}{2\pi} \tag{42}$$

$$Q(\omega) = |b_{0\omega}|^2 + \int |b_{k\omega}|^2 \frac{dk}{2\pi} \tag{43}$$

Midpoint Duhamel - Self conv. test results for $\mathrm{S}(\mathrm{x},\mathrm{t}) = \widehat{\mathrm{p}}\psi(\mathrm{x},\mathrm{t})$ - ZRP

$$\phi^{\omega}_{n+1} = U(\Delta t)\phi^{\omega}_{n} - i\Delta t e^{i\omega t_{n+\frac{1}{2}}} S_{n+\frac{1}{2}} \tag{38}$$

t	$\Re\,\psi(1)$	$\Im\,\psi(1)$	$\lVert\psi\rVert$	$\Re\,\phi(2)$	$\Im\,\phi(2)$	$\lVert\phi\rVert$
-1102.3132	1.0000000	0.0000000	1.0000000	0.0000000	1.4733146(-4)	0.6074294
-1080.2669	1.0000000	-1.5299150(-5)	1.0000000	9.0161878(-8)	1.4733146(-4)	2.9977570
-1058.2207	1.0000000	-3.0598299(-5)	1.0000000	1.8032375(-7)	1.4733145(-4)	4.2519438
			. . .			
-22.0463	-4.6878735(-7)	1.1266426(-6)	1.0000000	-3.4864304(-3)	-2.0577179(-3)	0.7712524
-0.0000	4.8643339(-1)	-9.9881227(-1)	1.0000000	-1.3514523	-8.0617898(-4)	3.3039060
22.0463	2.8495297(-7)	1.2366110(-6)	1.0000000	-3.7992762(-3)	1.3930728(-3)	4.2371529
			. . .			
1058.2207	-9.9526450(-1)	-9.7202282(-2)	1.0000000	1.5396368(-5)	-1.4659389(-4)	1.1029687
1080.2669	-9.9526599(-1)	-9.7187055(-2)	1.0000000	1.5389349(-5)	-1.4659255(-4)	3.5763755
1102.3132	-9.9526748(-1)	-9.7171829(-2)	1.0000000	1.5382713(-5)	-1.4659538(-4)	4.1762517

Level	Δt (relative)	E_{ψ}	E_{ϕ}	p_{ψ}/p_{ϕ}
1	Δt_0	0.1185025(-1)	0.1632911(+0)	—
2	$\Delta t_0/2$	0.2953778(-2)	0.3951342(-1)	2.004 / 2.048
3	$\Delta t_0/4$	0.7378959(-3)	0.9798974(-2)	2.001 / 2.011
Observed order				2.0011/2.0116

$$\mathrm{Ne} = 100, \quad \mathrm{Np} = 10$$

$$-\mathrm{xmin} = \mathrm{xrange}/2 = \mathrm{Np}\,\mathrm{Ns}/2$$

$$\tau = 2\pi n_{oc}/\omega_0,\ T = 8\tau$$

Initial conditions and parameters

$$S(x,t) = -i\partial_x\psi(x,t)$$

$$F(t) = F_0\, e^{-(2t'/\tau)^2}\cos(\omega_0 t),\ t' = t - T/2$$

$$F_0 = 0.03,\ \ \omega_0 = 0.114,\ \ \omega = 0.5$$

$$\Delta t_0 = 2.204,\ \ n_{oc} = 5$$

- Results of the time propagation for the zero range potential with and without source

$$p^{\mathrm{self}}_k = (\mathrm{err}_{k-1}/\mathrm{err}_k)\,/\log(2)$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Simpson Duhamel - Self conv. test results for $\mathrm{S}(\mathrm{x},\mathrm{t}) = \widehat{\mathrm{p}}\psi(\mathrm{x},\mathrm{t})$ - ZRP

$$\phi^{\omega}_{n+1} = U_4(\Delta t)\phi^{\omega}_n - \frac{i\Delta t}{6}\left(U_4(\Delta t)e^{i\omega t_n}S_n + 4U_4(\Delta t/2)e^{i\omega t_{n+\frac{1}{2}}}S_{n+\frac{1}{2}} + U_4(0)e^{i\omega t_{n+1}}S_{n+1}\right) \tag{38}$$

$$U_4(t) = U(c_1 h)U(c_2 h)U(c_1 h), \tag{35}$$

t	$\Re\,\psi(1)$	$\Im\,\psi(1)$	$\Vert\psi\Vert$	$\Re\,\phi(2)$	$\Im\,\phi(2)$	$\Vert\phi\Vert$
-1102.3132	1.0000000	0.0000000	1.0000000	0.0000000	1.4733146(-4)	0.6074294
-1080.2669	1.0000000	-1.5299150(-5)	1.0000000	9.0161878(-8)	1.4733146(-4)	2.9977570
-1058.2207	1.0000000	-3.0598299(-5)	1.0000000	1.8032375(-7)	1.4733145(-4)	4.2519438
			. . .			
-22.0463	-4.6878735(-7)	1.1266426(-6)	1.0000000	-3.4864304(-3)	-2.0577179(-3)	0.7712524
-0.0000	4.8643339(-1)	-9.9881227(-1)	1.0000000	-1.3514523	-8.0617898(-4)	3.3039060
22.0463	2.8495297(-7)	1.2366110(-6)	1.0000000	-3.7992762(-3)	1.3930728(-3)	4.2371529
			. . .			
1058.2207	-9.9526450(-1)	-9.7202282(-2)	1.0000000	1.5396368(-5)	-1.4659389(-4)	1.1029687
1080.2669	-9.9526599(-1)	-9.7187055(-2)	1.0000000	1.5389349(-5)	-1.4659255(-4)	3.5763755
1102.3132	-9.9526748(-1)	-9.7171829(-2)	1.0000000	1.5382713(-5)	-1.4659538(-4)	4.1762517

Level	Δt (relative)	E_ψ	E_ϕ	p_ψ/p_ϕ
1	Δt_0	0.5564105(-3)	0.1529880(-1)	—
2	$\Delta t_0/2$	0.3480484(-4)	0.1093299(-2)	3.999 / 3.807
3	$\Delta t_0/4$	0.2175758(-5)	0.7047939(-4)	4.000 / 3.955
Observed order				3.9997/3.9553

$$\mathrm{Ne} = 100, \quad \mathrm{Np} = 10$$

$$-\mathrm{xmin} = \mathrm{xrange}/2 = \mathrm{Np}\,\mathrm{Ns}/2$$

$$\tau = 2\pi n_{oc}/\omega_0, \; T = 8\tau$$

Initial conditions and parameters

$$S(x,t) = -i\partial_x\psi(x,t)$$

$$F(t) = F_0\, e^{-(2t'/\tau)^2}\cos(\omega_0 t), \; t' = t - T/2$$

$$F_0 = 0.03, \;\; \omega_0 = 0.114, \;\; \omega = 0.5$$

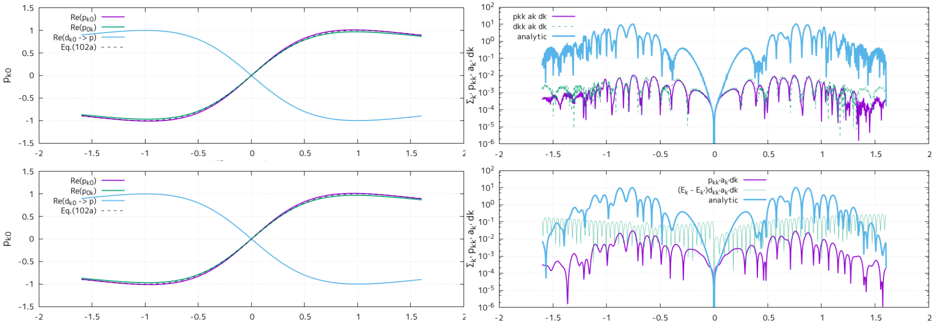
$$\Delta t_0 = 2.204, \;\; n_{oc} = 5$$

- Results of the time propagation for the zero range potential with and without source

$$p^{\mathrm{self}}_k = (\mathrm{err}_{k-1}/\mathrm{err}_k)\,/\log(2)$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Transition amplitudes



$$\mathrm{Np} = 20, \quad \mathrm{Ns} = 100$$
$$\mathrm{xmax} \sim \mathrm{Np}\,\mathrm{Ns}/2$$
$$\omega_0 = 0.114\,\mathrm{a.u}$$
$$F_0 = 0.03\,\mathrm{a.u}, \quad T = 8\tau$$

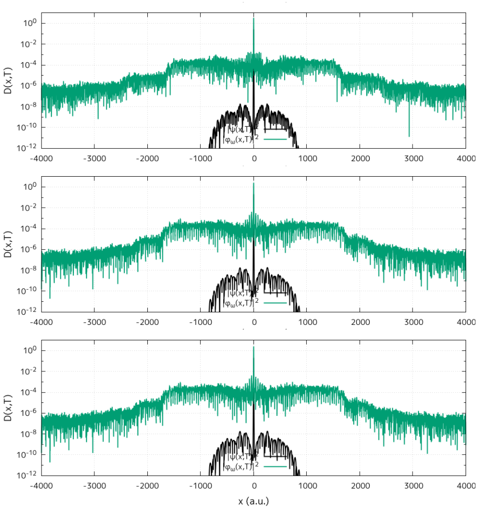
$$p_{k0} = \int dx\, \psi_k^{(-)*}(x)\, \hat{p}\, \psi_0(x) \tag{44}$$

U: $\mathrm{Np} = 20, \mathrm{Ns} = 100,\ \mathrm{xmax} = 1000$
D: $\mathrm{Np} = 40, \mathrm{Ns} = 200,\ \mathrm{xmax} = 4000$

$$p_{kk'}\, a_k\, dk = \left(\int dx\, \psi_k^{(-)*}(x)\, \hat{p}\, \psi_{k'}^{(-)}(x) \right) a_k\, dk \tag{45}$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Convergence region for ϕ_ω in x, $T = 5\tau$, with $F_0 = 0.03\,\mathrm{a.u.}$ and $\omega_0 = .114\,\mathrm{a.u.}$



1st row $\quad \mathrm{Np} = 20, \quad \mathrm{Ns} = 300$

3rd row $\quad \mathrm{nsteps} = 20000$

other rows $\quad \mathrm{Np} = 30, \quad \mathrm{Ns} = 200$

other rows $\quad \mathrm{nsteps} = 15000$

$\mathrm{xmax} \sim \mathrm{Np}\,\mathrm{Ns}(1+\mathrm{q})/2,$ **Same row, same info**

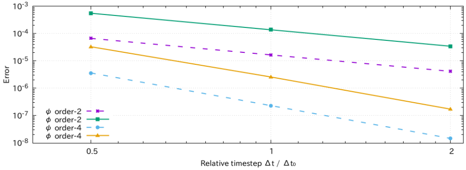
Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Quantum High Harmonic generation, for ϕ_ω with $\omega = 0.5\,\mathrm{a.u.}$

$b_{0\omega}$	$\mathrm{Re}(b_{0\omega})$	$\mathrm{Im}(b_{0\omega})$	$\lvert b_{0\omega}\rvert^2$	$\mathrm{Re}(Q(\omega))$	$\mathrm{Im}(Q(\omega))$
3.1399646834322419	1.7620682503078942	0.0	12.964262731944862	12.964262731944862	0.0

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Self conv. test results for real tests, both orders



$$E \sim C(\Delta t)^p$$
$$\log E = \log C + p \log(\Delta t)$$
$$C_2 \sim 1.66(-5)$$
$$C_4 \sim 2.31(-7)$$

Level	Δt (relative)	E_ψ	E_ϕ	p_ψ / p_ϕ
1	$2\Delta t_0$	0.6673536(-4)	0.5532128(-3)	—
2	Δt_0	0.1661859(-4)	0.1378364(-3)	2.006 / 2.005
3	$\Delta t_0/2$	0.4150871(-5)	0.3445575(-4)	2.001 / 2.000
Observed order				2.0013/2.0001

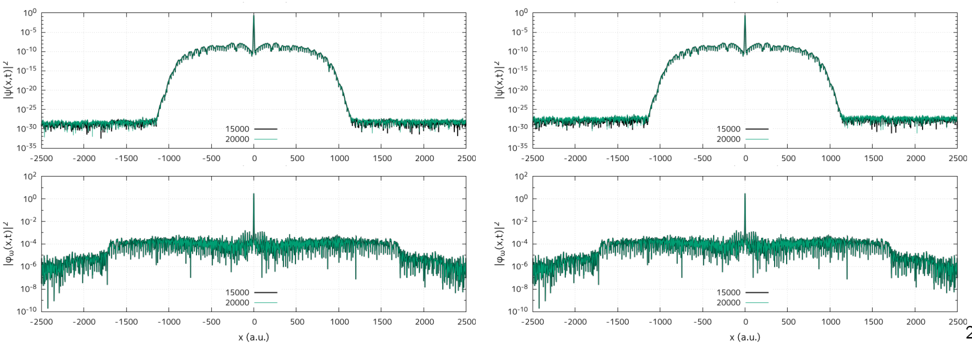
Level	Δt (relative)	E_ψ	E_ϕ	p_ψ / p_ϕ
1	$2\Delta t_0$	0.3627616(-5)	0.3289495(-4)	—
2	Δt_0	0.2322254(-6)	0.2587140(-5)	3.965 / 3.668
3	$\Delta t_0/2$	0.1460063(-7)	0.1720141(-6)	3.991 / 3.910
Observed order				3.9914/3.9108

$$\mathrm{Np} = 20, \quad \mathrm{Ns} = 100$$
$$\mathrm{xmax} \sim \mathrm{Np}\,\mathrm{Ns}(1 + 1/4)/2$$
$$\omega_0 = 0.114\,\mathrm{a.u}$$
$$F_0 = 0.03\,\mathrm{a.u}, \quad T = 5\tau$$
$$\mathrm{nsteps} = 20000, \quad \Delta t_0 = 0.0689$$

$$p_k^{\mathrm{self}} = (\mathrm{err}_{k-1}/\mathrm{err}_k)\,/\log(2)$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Δt Time step convergence for the inhomogeneous problem



1st column : order 2

2nd column : order 4

$$\mathrm{Np} = 20, \quad \mathrm{Ns} = 200$$

$$\mathrm{xmax} \sim \mathrm{Np}\,\mathrm{Ns}(1 + 1/4)/2$$

$$\omega_0 = 0.114\,\mathrm{a.u}$$

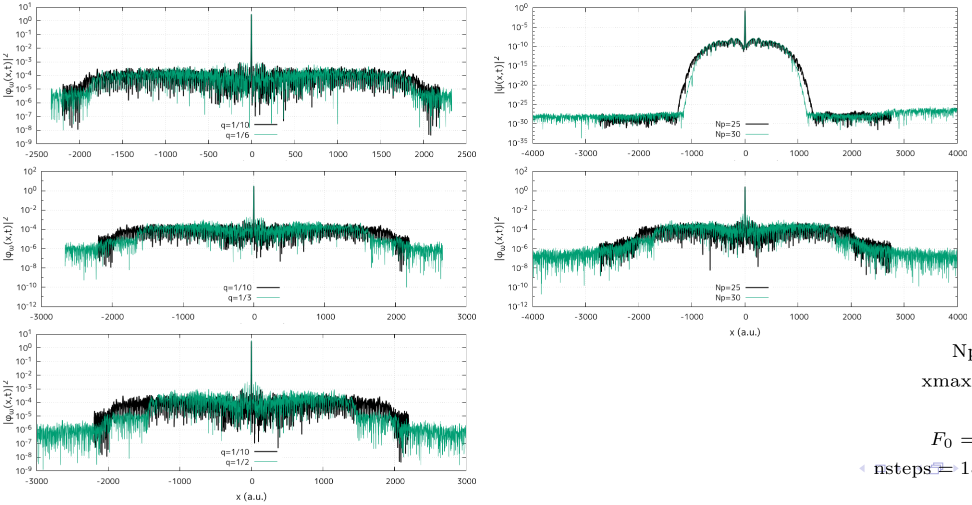
$$F_0 = 0.03\,\mathrm{a.u}, \quad T = 5\tau$$

$$\mathrm{nsteps} = 15000, \quad \Delta t = 0.092$$

- No improvement brought by the order 4 scheme whatsoever
- In principle, order 4 should require less steps, i.e tolerate a
- $\Delta t \leqslant 0.08$ sufficient for convergence for the inhomogeneous problem

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Space grid convergence for the inhomogeneous problem



$$\mathrm{Np} = 20, \quad \mathrm{Ns} = 200$$

$$\mathrm{xmax} \sim \mathrm{Np\,Ns}(1+\mathrm{q})/2$$

$$\omega_0 = 0.114\,\mathrm{a.u}$$

$$F_0 = 0.03\,\mathrm{a.u}, \quad T = 5\tau$$

$$\mathrm{nsteps} = 15000, \quad \Delta t = 0.092$$